

130 nm RGC-CD TIA Design Doc

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1 Introduction

This report summarises the design of a transimpedance amplifier (TIA) I developed as a member of the University of Waterloo’s ASIC student design team during my Summer 2025 co-op semester. The TIA is the current-to-voltage front-end of a larger analog matrix-vector multiplier (MVM). In that system the multiplication is carried out by weighted current mirrors, so every column of a matrix $\mathbf{A}_{n \times n}$ produces an output current that must be converted to a voltage before it can be digitised by an analog-to-digital converter (ADC).

Because an $n \times n$ MVM requires n such converters, silicon area is the dominant constraint. Providing each column with its own compact converter avoids the bottleneck of a single centralised TIA fronted by an n -input current multiplexer. This motivated a regulated-cascode (RGC) input stage rather than an operational-amplifier based TIA: the RGC presents a low input impedance for current sensing while using far fewer devices (and hence less area) than an op-amp design. The RGC stage is followed by a capacitive-degeneration (CD) stage that extends the closed bandwidth. The overall topology is adapted from the low-power TIA of [1]. The complete schematic, with the RGC input stage and CD stage highlighted, is shown in Figure 1.

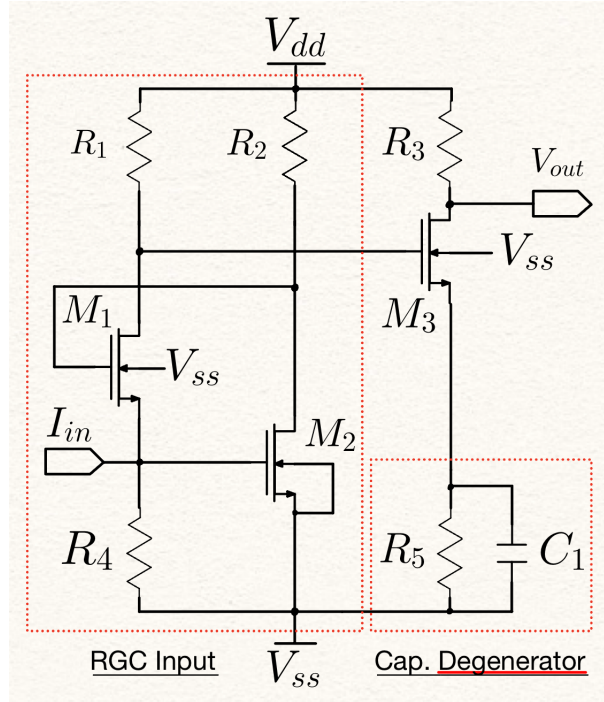


Figure 1: Complete RGC-CD TIA schematic (RGC input stage + capacitive-degeneration stage).

The entire design was carried out with an open-source flow: `xschem` for schematic capture, `ngspice` for simulation, and `magic/klayout` for layout. The topology of Figure 1 gives rise to the following net

identities, used without further comment throughout:

$$V_{S_1} = V_{G_2} \quad (1)$$

$$V_{G_1} = V_{D_2} \quad (2)$$

$$V_{D_1} = V_{G_3} \quad (3)$$

$$V_{out} = V_{D_3} \quad (4)$$

As discussed in Section 3.6, the single-ended output biases this topology asymmetrically and limits it to a largely unidirectional (single-sign) input current which works well with the current-mirror MVM front-end.

2 Prior Work: The Original Baseline

The circuit of Figure 1 was first designed and simulated against the SKY130B 0.13 μm CMOS process to comply with the open-source TinyTapeout flow but never reached the layout stage. The present document describes a subsequent revitalisation of that design on the SKY130A device models. The topology is identical between the two PDKs, only the sizing methodology and the device models differ. For context, the headline results of the original SKY130B baseline are collected in Table 1.

Table 1: Original SKY130B baseline (identical topology, pre-layout, TT, 27°C).

Parameter	Value	Units
Passive values (R_1, R_2, R_4)	2	$k\Omega$
Passive values (R_3, R_5)	6, 5.895	$k\Omega$
Degeneration cap C_1	450	fF
Full-TIA DC gain	58.9 (−884)	$dB\Omega$ (Ω)
RGC-only gain	61.7	$dB\Omega$
Peaking	~ 0.75	dB
RGC-only cutoff f_{-3dB}	66	MHz
RGC+CD cutoff f_{-3dB}	143	MHz
Power	1.06	mW
Input signal range	−5 to +20	μA

The original sizing followed an overdrive-voltage (V_{ov}) approach: every device was sized from the square-law relation $I_D = \frac{1}{2}k'\frac{W}{L}V_{ov}^2$ with a fixed $V_{ov} = 0.2\text{ V}$, an assumed $V_{th} = 0.4\text{ V}$ (LVT), and an assumed process transconductance $k' = 60\text{ }\mu A/V^2$. This is simple to hand-calculate, but it assumes strong-inversion square-law behaviour and a transconductance parameter that is not, in practice, well known before characterisation. On the faster SKY130A models the assumed k' proved inaccurate (the measured RGC gain landed on the hand-calculated $\sim 66\text{ dB}\Omega$ rather than the $61.7\text{ dB}\Omega$ originally simulated), and the frequency-dependent quantities did not port at all: the intrinsic RGC pole moved from the anticipated 66 MHz up to $\approx 2\text{ GHz}$.

For the revitalisation the design was therefore re-derived from the ground up using a g_m/I_D methodology, in which every device is sized from measured current-density and speed data rather than from an assumed square-law constant. The full g_m/I_D design flow, small-signal analysis, and SKY130A simulation results are the subject of Section 3.

3 RGC-CD Design Flow (g_m/I_D , SKY130A)

3.1 g_m/I_D Design Methodology

The g_m/I_D method sizes each transistor from measured device curves rather than from an assumed square-law k' . A unit-width device is swept over V_{GS} at a fixed $|V_{DS}| = V_{DD}/2$, and per operating point the small-signal quantities g_m , g_{ds} , C_{gg} are extracted directly from the BSIM instance inside the SKY130 model. Because g_m/I_D and the current density $J_D = I_D/W$ are essentially width-independent, a single unit device characterises the whole family.

The operating-point choice trades efficiency against speed. A high g_m/I_D (weak/moderate inversion) maximises transconductance per unit current and intrinsic gain $g_m r_o$ but gives a low transit frequency

f_T . A low g_m/I_D (strong inversion) does the opposite. Figure 2 shows the four master charts for the LVT nfet used throughout the RGC-CD signal path, and Figure 3 overlays the nfet against the pfet at the channel lengths the candidate topologies actually use.

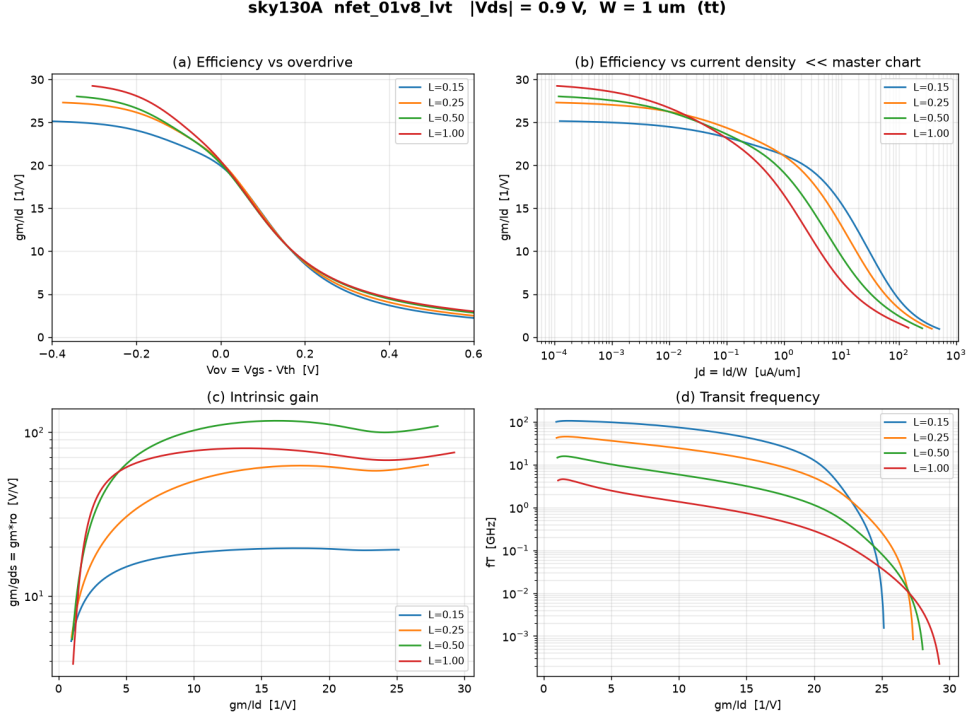


Figure 2: g_m/I_D master charts for the SKY130A LVT nfet ($|V_{DS}| = 0.9\text{ V}$): (a) efficiency vs. overdrive, (b) efficiency vs. current density J_D (the master sizing chart), (c) intrinsic gain $g_m r_o$, and (d) transit frequency f_T , each vs. g_m/I_D .

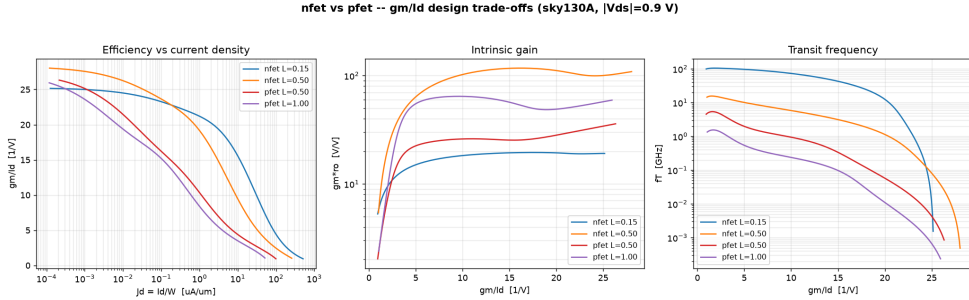


Figure 3: Cross-device g_m/I_D trade-offs (efficiency, intrinsic gain, f_T) for the nfet and pfet at the lengths the RGC-CD front-end uses.

The cascode/auxiliary devices M_1, M_2 were placed at $g_m/I_D = 18\text{ V}^{-1}$ (moderate inversion, a balance of loop gain and speed), and the degeneration device M_3 at $g_m/I_D = 22\text{ V}^{-1}$ (slightly more efficient, since its speed is set by the degeneration network rather than by f_T). Reading J_D , $g_m r_o$, and f_T off the charts at the chosen bias currents yields the sizing summarised in Table 2. All devices use the minimum $L = 0.15\text{ }\mu\text{m}$. Each is to be laid out multi-finger for implementation, with negligible electrical effect (Section 3.8).

Table 2: g_m/I_D device sizing (SKY130A LVT nfet, $L = 0.15 \mu\text{m}$).

Device	I_D [μA]	g_m/I_D [V^{-1}]	W [μm]	g_m [mS]	$g_m r_o$	f_T [GHz]
M_1 (cascode)	60	18	7.3	1.08	14.4	36.0
M_2 (aux CS)	55	18	7.9	0.99	14.9	29.8
M_3 (degeneration)	40	22	19.1	0.88	14.3	12.2

The passive network that biases this sizing (and shapes the response of Section 3.2) is $R_1 = 6.5 k\Omega$, $R_2 = 6 k\Omega$, $R_3 = 4 k\Omega$, $R_4 = 9 k\Omega$, $R_5 = 22.5 k\Omega$, and $C_1 = 178 fF$. The front-end is deliberately current-starved: a large R_5 with a low I_3 maximises $g_{m3}R_5$, which (as shown below) pushes the degeneration pole high, while $R_4 = 9 k\Omega$ keeps the input node biased for adequate saturation margin.

3.2 Small-Signal Transfer Function and Pole/Zero Set

3.2.1 RGC front-end

The auxiliary common-source amplifier M_2 ($V_{G_2} = V_{S_1}$, load R_2) provides a regulating gain

$$A_{aux} = g_{m2}R_2 \approx 5.9 V/V. \quad (5)$$

This gain boosts the shunt feedback around M_1 , lowering the input impedance seen by the injected signal current to

$$Z_{in} = \frac{1}{g_{m1}(1 + A_{aux})} \parallel R_4 \approx 132 \Omega. \quad (6)$$

The signal current flows through M_1 into R_1 , so the front-end transimpedance to the RGC output node V_{D_1} is set (to first order) by the load resistor, $Z_{T,\text{RGC}} \approx R_1$.

3.2.2 Capacitive-degeneration stage

The CD stage is a common-source amplifier M_3 whose source is degenerated by the parallel network $Z_5(s) = R_5 \parallel (1/sC_1) = R_5/(1 + sR_5C_1)$. Writing KCL at V_{out} and V_{S_3} (small-signal model in Figure 4),

$$\frac{V_{out} - V_{S_3}}{r_{o3}} + \frac{V_{out}}{R_3} + g_{m3}(V_{G_3} - V_{S_3}) = 0 \quad (7)$$

$$-g_{m3}(V_{G_3} - V_{S_3}) + \frac{V_{S_3} - V_{out}}{r_{o3}} + \frac{V_{S_3}}{Z_5(s)} = 0 \quad (8)$$

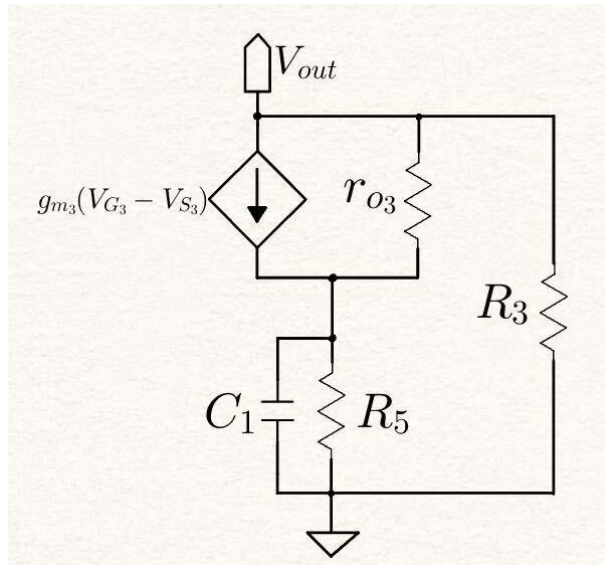


Figure 4: Small-signal model of the capacitive-degeneration stage (M_3 degenerated by $R_5 \parallel C_1$).

Eliminating V_{S_3} gives a single-zero, single-pole response

$$A_{CD}(s) = A_0 \frac{1 + s/\omega_z}{1 + s/\omega_p}, \quad A_0 = \frac{g_{m3}R_3}{1 + g_{m3}R_5}, \quad (9)$$

$$\omega_z = \frac{1}{R_5C_1}, \quad \omega_p = \frac{1 + g_{m3}R_5}{R_5C_1} \approx \frac{g_{m3}}{C_1}. \quad (10)$$

Because $g_{m3}R_5 \gg 1$, the pole sits well above the zero: the stage produces a rising (peaking) response between ω_z and ω_p that is used to extend the closed bandwidth. The CD stage is a common-source amplifier, so A_0 is in fact negative: it both attenuates ($|A_0| \approx 0.17$, the source degeneration $g_{m3}R_5$ dividing the gain down) and inverts the overall TIA output. The load-resistor front-end contributes the transimpedance magnitude, the common-source stage its sign. The positive A_0 written above is thus a magnitude, as is the $Z_{T,DC}$ term below.

3.2.3 Loaded transfer function

In the deployed system the input node carries the MVM current-mirror capacitance ($C_T \approx 500 \text{ fF}$) and the output drives an ADC sampling input ($C_L \approx 1 \text{ pF}$). These introduce two additional load-induced poles, one at the input node and one at the output node:

$$\omega_{in} = \frac{1}{Z_{in}C_T}, \quad \omega_{out} = \frac{1}{R_3C_L}. \quad (11)$$

Collecting the front-end, CD stage, internal nodes and loads, the full first-order transimpedance is

$$Z_T(s) = R_1 \underbrace{\frac{g_{m3}R_3}{1 + g_{m3}R_5}}_{Z_{T,DC}} \cdot \frac{1 + s/\omega_z}{\left(1 + \frac{s}{\omega_{in}}\right) \left(1 + \frac{s}{\omega_{aux}}\right) \left(1 + \frac{s}{\omega_{D1}}\right) \left(1 + \frac{s}{\omega_p}\right) \left(1 + \frac{s}{\omega_{out}}\right)}. \quad (12)$$

Evaluating each singularity at the sizing of Table 2 gives the pole/zero set of Table 3. The internal-node poles ω_{D1} (transimpedance node) and ω_{aux} (aux-loop node) are gate-capacitance-dominated estimates.

Table 3: First-order pole/zero set of the loaded transimpedance (g_m/I_D design, $C_T = 500 \text{ fF}$, $C_L = 1 \text{ pF}$).

Singularity	Type	Frequency	Expression
f_z (CD zero)	zero	40 MHz	$1/(2\pi R_5C_1)$
f_{out} (output/load)	pole	40 MHz	$1/(2\pi R_3C_L)$
f_p (CD pole)	pole	827 MHz	$(1 + g_{m3}R_5)/(2\pi R_5C_1)$
f_{D1} (transimp. node)	pole	2.13 GHz	$1/(2\pi R_1C_{D1})$
f_{in} (input/load)	pole	2.42 GHz	$1/(2\pi Z_{in}C_T)$
f_{aux} (aux loop)	pole	5.56 GHz	$1/(2\pi R_2C_{D2})$

3.3 s-Plane View: Zero–Pole Cancellation

Every singularity in Table 3 is real and lies in the left-half plane, so the s -plane collapses onto the negative real axis (Figure 5). The value of the CD stage is now explicit. Without any degeneration capacitor the response would roll off at the lowest pole, which is the output/load pole $f_{out} = 1/(2\pi R_3C_L) \approx 40 \text{ MHz}$ set by the 1 pF ADC load. The degeneration capacitor is sized so that the CD zero lands on that load pole,

$$R_5C_1 = R_3C_L \implies f_z = f_{out} = 40 \text{ MHz}, \quad (13)$$

which cancels it. With the load pole removed, the dominant pole role passes to the CD pole $f_p \approx 827 \text{ MHz}$ and the $\sim 2 \text{ GHz}$ intrinsic cluster (f_{D1}, f_{in}), so the closed bandwidth is extended by more than a decade. The input-load pole f_{in} and the aux-loop pole f_{aux} sit even higher and do not limit the response.

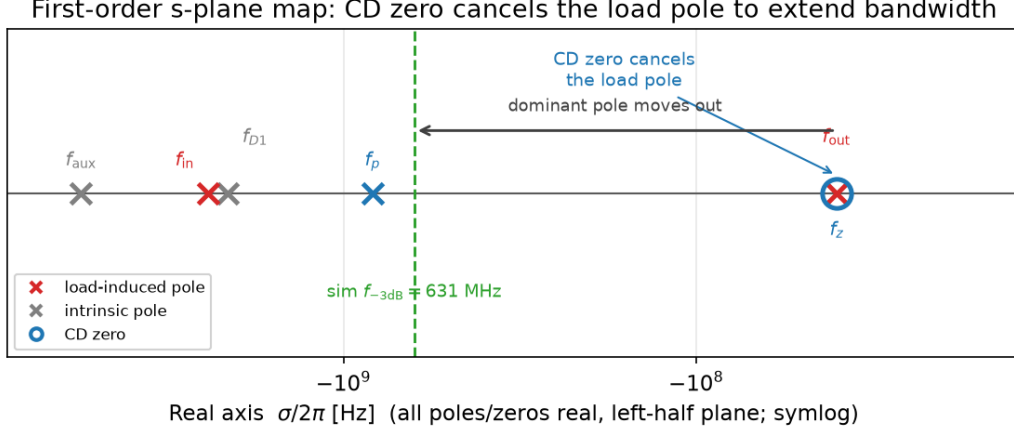


Figure 5: First-order s -plane map of the loaded transimpedance. The CD zero f_z is placed on the output (load) pole f_{out} . Cancelling it moves the dominant pole out to the $\sim$ GHz cluster, extending the bandwidth to the simulated 631 MHz.

3.4 Simulated DC and Frequency Response

The g_m/I_D sizing was verified in ngspice on the SKY130A models. Every device settles in saturation. The operating-point currents drift modestly from the sizing targets ($I_1, I_2, I_3 \approx 68/80/33 \mu A$ vs. $60/55/40 \mu A$ targeted), giving a total supply current of $0.18 mA$ and a power of

$$P_{TIA} = V_{DD} \cdot I_{tot} = (1.8 V)(0.18 mA) \approx 0.33 mW, \quad (14)$$

a $3\times$ reduction from the original $1.06 mW$ baseline. The simulated DC transimpedance is $59.4 dB\Omega$ (933Ω), with the output parked high at a nominal $V_{out} \approx 1.67 V$ (see Section 3.6).

Figure 6 isolates the capacitive degenerator by comparing the loaded response with and without C_1 . With only resistive degeneration ($C_1 = 0$) the transimpedance rolls off at the bare output/load pole at $\approx 39 MHz$. Restoring C_1 cancels that pole and extends the $-3 dB$ bandwidth to 631 MHz, a $\sim 16\times$ extension at unchanged DC gain and with negligible ($< 0.1 dB$) peaking. This is exactly the zero–pole cancellation of Section 3.3. The remaining two overlays in the figure, the folded real-device netlist and the parasitic-extracted layout, preview the physical implementation of Section 4. Both preserve this flat frequency response with only a modest bandwidth trim.

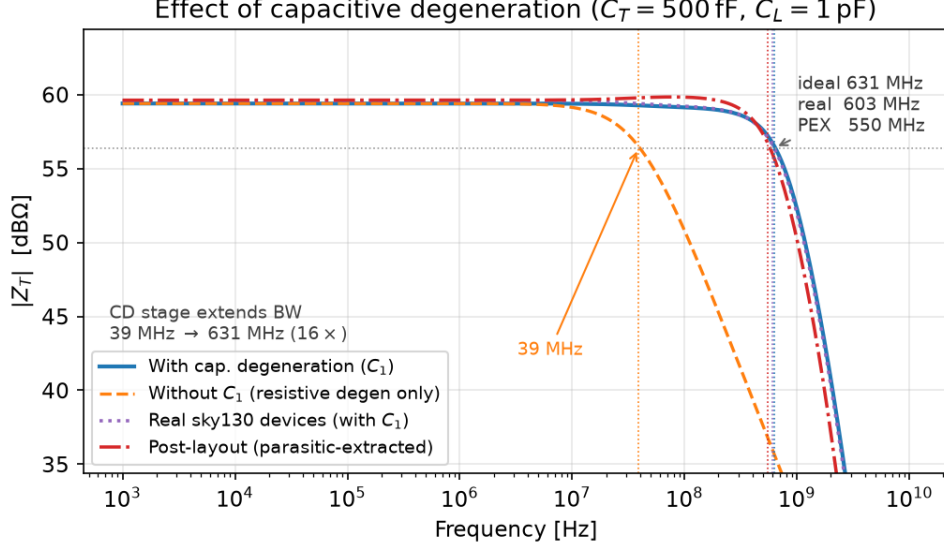


Figure 6: Effect of the capacitive-degeneration stage on the loaded transimpedance (SKY130A, TT, 27°C, $C_T = 500\text{ fF}$ input, $C_L = 1\text{ pF}$ ADC load). The ideal single-finger design is shown with and without C_1 . Overlaid are the folded real-device and post-layout parasitic-extracted responses of Section 4, which retain the flat DC gain while trimming the -3 dB bandwidth from 631 to 603 and 556 MHz respectively.

It is worth stressing that this compensation is load- and process-dependent: the CD zero must be placed on whichever pole actually dominates (here the ADC-load pole), and re-centred if the target load or bias changes. This is precisely why the original $C_1 = 450\text{ fF}$ (sized on SKY130B for a 66 MHz pole) over-peaks on the faster SKY130A devices, and why C_1 is exposed as a tunable parameter in the reconstructed netlist. In an unloaded testbench, with the RGC pole near 2 GHz and no load pole for the zero to cancel, the same C_1 produces a large ($\sim +17\text{ dB}$) resonant peak.

3.5 Loop Stability

Because the RGC relies on the local M_1 – M_2 regulation loop, its stability must be checked at the deployed operating point. Figure 7 shows the loop gain and phase under the same C_T/C_L loading. The loop has a DC gain of $\approx 12\text{ dB}$ and crosses unity gain near 940 MHz with $\approx 130^\circ$ of phase margin, so the flat extended response of Section 3.4 is a well-damped operating point rather than a lightly-damped resonance.

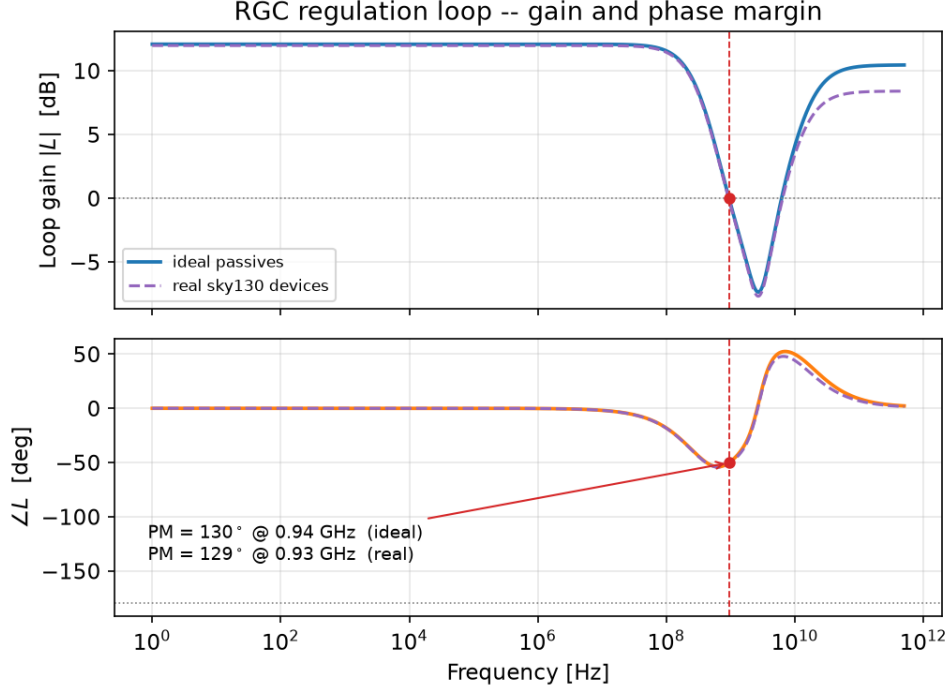


Figure 7: RGC regulation-loop gain and phase margin (SKY130A, TT, 27°C, with $C_T = 500 \text{ fF}$ input and $C_L = 1 \text{ pF}$ load).

3.6 Input Linearity and the Negative-Current Limitation

The input current is injected at the source of M_1 , a low-impedance node ($Z_{in} \approx 132 \Omega$) held near 0.6 V by the auxiliary loop, while the single-ended output is biased above mid-rail ($\sim 1.67 \text{ V}$). The available output swing is therefore asymmetric: a positive (sourcing) input pulls V_{out} down toward V_{SS} (ample headroom), whereas a negative input pushes V_{out} up toward V_{DD} (limited headroom). At the system level a current-mirror MVM front-end can only source current in one direction, so realising genuinely bidirectional (signed) input currents would require additional current-steering circuitry.

The core amplifier is more robust than the original $-5/+20 \mu\text{A}$ signal window suggests. The DC linearity sweep of Figure 8 shows the small-signal transimpedance staying within 10% of its nominal value over roughly -85 to $+65 \mu\text{A}$, with hard clipping only beyond those points (as $V_{out} \rightarrow V_{DD}$ for large negative input, and $V_{out} \rightarrow V_{SS}$ for large positive input). The original $-5/+20 \mu\text{A}$ figures thus represent the signal window chosen to match the expected (largely unidirectional) MVM currents rather than a hard electrical limit of the core.

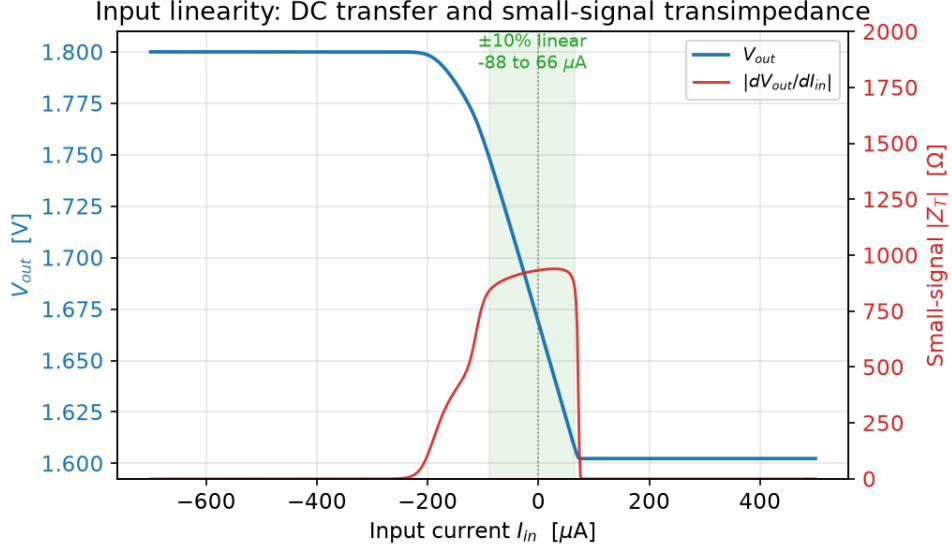


Figure 8: Input linearity: DC transfer $V_{out}(I_{in})$ and small-signal transimpedance vs. operating point (SKY130A, TT, 27°C).

3.7 Input-Referred Noise

The input-referred current noise is obtained by integrating the input-referred noise spectral density over the design band. Integrating over the loaded band (1 kHz to the 631 MHz -3 dB point of Section 3.4) gives

$$\bar{I}_{n,rms}|_{T=27^\circ C} = \sqrt{\int_{1\text{ kHz}}^{631\text{ MHz}} i_n^2(f) df} \approx 158\text{ nA}_{rms}. \quad (15)$$

This is substantially larger than the $\sim 73\text{ nA}_{rms}$ obtained over the old $\sim 140\text{ MHz}$ band, for two compounding reasons: the band is more than $4\times$ wider, and the input-referred density rises at high frequency as the transimpedance $|Z_T|$ rolls off (input-referred noise = output noise / $|Z_T|$). The spectral density is shown in Figure 9: a $1/f$ (flicker) roll-off below $\sim 20\text{ kHz}$ down to a white thermal floor of $\approx 9.6\text{ pA}/\sqrt{\text{Hz}}$, then the high-frequency rise from the transimpedance roll-off. The temperature dependence is given in Table 4. Unlike the baseline design, integrating to a fixed 631 MHz makes the total noise increase with temperature, as expected for a thermally-dominated floor.

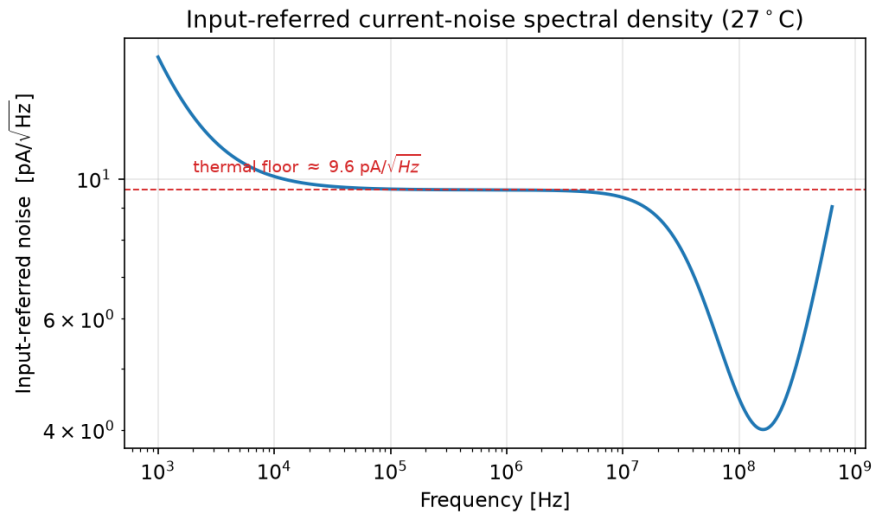


Figure 9: Input-referred current-noise spectral density (SKY130A, TT, 27°C, loaded).

Table 4: Input-referred noise vs. temperature (integrated 1 kHz–631 MHz, loaded).

Temp [°C]	$\bar{I}_{n,rms}$ [nA_{rms}]	$V_{o,n}$ [μV_{rms}]
27	158.2	127.8
45	165.3	131.7
75	177.1	137.8

3.8 Device Fingering for Layout

For physical implementation the signal devices will be drawn multi-finger: M_1 , M_2 , and the wide degeneration device M_3 each with $n_f = 4$ fingers (per-finger widths of 1.8, 2.0 and 4.8 μm). The electrical design of record above is characterised at $n_f = 1$. Folding is a layout choice, not an electrical one: the total gate width, and hence g_m , I_D , and every gain and pole of Table 2, is unchanged. The motivation is purely physical: a single-finger M_3 is a 19.1 $\mu\text{m} \times 0.15 \mu\text{m}$ sliver (127:1 aspect ratio) that is awkward to place and route, whereas four 4.8 μm fingers form a compact cell whose adjacent fingers share source/drain diffusions. Folding also enables interdigitated / common-centroid placement of the matched M_1 – M_2 regulation pair.

Re-simulating the folded netlist confirms the change is electrically negligible (Table 5). The redistributed source/drain junction geometry nets a small increase in internal-node parasitic capacitance, which lowers the loaded bandwidth by $\sim 4\%$ (631 $\rightarrow$ 603 MHz) and, at a *common* integration band, raises the input-referred noise by $\sim 2.5\%$. DC gain, phase margin and power are unchanged to rounding. Integrated to the folded variant’s own 603 MHz -3dB point, its input-referred noise would be 154.5 nA_{rms} .

Two caveats frame this result. First, the pre-layout SKY130 device models do *not* include distributed gate resistance (r_{gate} is disabled), so the principal benefit of fingering (the $\propto 1/n_f^2$ collapse of gate-resistance thermal noise, which for a single-finger M_3 would otherwise rival its channel noise) does not appear in these numbers. The cap-only parasitic extraction of Section 4.5 likewise omits it, so capturing that benefit will require a full RC extraction. Second, the small bandwidth/noise penalty seen here sits well within the SKY130A-vs-B model spread already noted in Section 2. Fingering is therefore adopted as electrically free pre-layout and net-beneficial after extraction.

Table 5: Effect of device fingering on the loaded design (SKY130A, TT, 27°C).

Metric	Single-finger ($n_f=1$)	Folded ($n_f=4$)
DC transimpedance [$dB\Omega$]	59.4	59.6
Loaded f_{-3dB}	631 MHz	603 MHz
Peaking [dB]	< 0.1	< 0.1
Loop phase margin	130°	129°
Input-referred noise [†] [nA_{rms}]	158.2	162.1
Power [mW]	0.33	0.33

[†] integrated over a *common* 1 kHz–631 MHz band to isolate the device effect. Over each design’s own -3dB band the folded figure is 154.5 nA_{rms} .

4 Physical Implementation

With the pre-layout design frozen, the TIA was carried through schematic capture, physical layout, layout-versus-schematic (LVS) verification, and parasitic extraction (PEX) on the open-source SKY130A flow inside the IIC-OSIC-TOOLS container. This section documents that back-end flow and quantifies the loaded gain and bandwidth once real devices and layout parasitics are accounted for.

4.1 Schematic Capture (xschem)

The design of record was entered in `xschem` to serve as the single source for both pre-layout simulation and the LVS reference. Figure 10 shows the captured schematic. The three signal devices M_1 – M_3 are `nfet_01v8_lvt` instances drawn multi-finger ($n_f = 4/4/4$) at the widths of Table 2. The bias network uses `res_high_po_0p35` for R_1 – R_4 and the higher-sheet `res_xhigh_po_0p69` for the large R_5 (design target 22.5 $k\Omega$, $\approx$ 22.8 $k\Omega$ as drawn). The degeneration capacitor C_1 is a `cap_mim_m3_1` MIM (17 $\times$ 5 μm , 178 fF). Net labels follow the identities of the Introduction ($d_1 = V_{G_3}$, $d_2 = V_{G_1}$, $s_3 = V_{S_3}$), and the

ports are I_{in} , V_{out} , V_{DD} , V_{SS} . Two dummy resistors (R_{dum1} , R_{dum2} , tied to V_{SS}) flank the active bank so the matched resistors see identical neighbours.

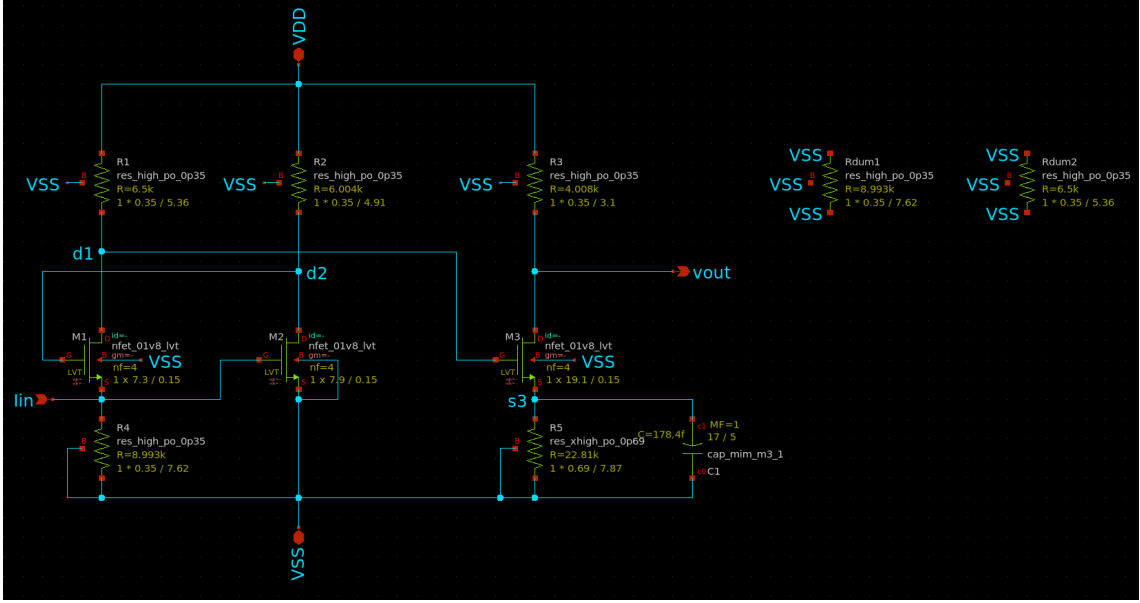


Figure 10: RGC-CD TIA captured in `xschem`: signal devices M_1 – M_3 (`nfet_01v8_lvt`, $n_f = 4/4/4$), the R_1 – R_5 poly-resistor bias network with two flanking dummies, and the MIM degeneration capacitor C_1 . This schematic drives both the pre-layout simulation and the LVS comparison.

4.2 Layout (magic)

The cell was laid out in `magic` on the SKY130A PDK (Figure 11). The three signal transistors are folded ($n_f = 4/4/4$, Section 3.8) with interdigitated fingers sharing source/drain diffusions. The poly resistors form a matched bank, the MIM degeneration capacitor sits alongside, and a substrate guard ring encloses the whole cell to fix the well/substrate potential. The completed core measures $25.46 \mu\text{m} \times 9.8 \mu\text{m}$ ($\approx 250 \mu\text{m}^2$), compact enough to be replicated per MVM column, which was the original area motivation for the RGC front-end (Section 2).

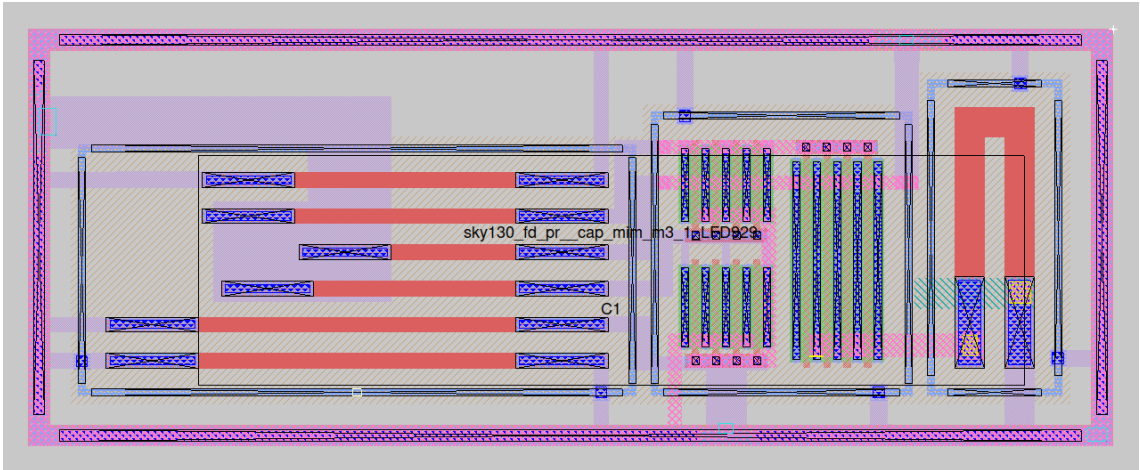


Figure 11: Finished `magic` layout of the RGC-CD TIA ($25.46 \times 9.8 \mu\text{m}$): the folded M_1 – M_3 transistor array, the matched poly-resistor bias bank, and the MIM degeneration capacitor, all enclosed by a substrate guard ring.

4.3 Dummy Resistors and Bank Matching

The bias resistors R_1 – R_4 set the operating point and the pole/zero locations, so their *ratios* matter more than their absolute values. To keep those ratios accurate the four active resistors are drawn as a single matched bank of parallel `res_high_po_0p35` strips on a common orientation and pitch. The dominant systematic error such a bank has to reject is the *edge effect*: the outermost strip in an array etches and planarises differently from an interior strip, because it sees open field on one side instead of an identical neighbour. Left uncorrected this skews the two edge resistors and breaks the very ratios the bank exists to hold.

The two dummy resistors `Rdum1` and `Rdum2` exist solely to absorb that edge error. They flank the bank on either side (bank order, bottom to top: `Rdum1`, R_4 , R_2 , R_3 , R_1 , `Rdum2`) so that every *active* resistor is bracketed by identical poly neighbours on both sides and no signal-setting device sits on the array edge. The dummies carry no signal (both terminals of each are tied to `VSS`), so they are electrically inert and appear in the netlist only as grounded devices. Their entire purpose is geometric, to reproduce the local environment (spacing, poly density, well proximity) that the interior strips enjoy.

For the matching to work each dummy need only replicate the environment its *nearest active neighbour* sees, which means matching that neighbour’s length rather than being made arbitrarily tall. This is also a manufacturability requirement, not just a nicety. In SKY130 the resistor mask (`rpm`, GDS 86/20) is *generated* from the poly-resistor bodies at GDS write and merges across adjacent strips of similar height. A dummy left taller than its short neighbour leaves an isolated `rpm` finger narrower than the $1.27\ \mu\text{m}$ minimum, which fails the `rpm.1a` width rule. Length-matching each dummy to its neighbour, `Rdum1` to R_4 ($L = 7.62\ \mu\text{m}$) and `Rdum2` to R_1 ($L = 5.36\ \mu\text{m}$), removes that protruding finger and closes the DRC at its source, while still presenting each edge resistor with a full-length matched neighbour. The result is clean sign-off DRC with the bank’s matching integrity intact.

4.4 Layout-vs-Schematic (LVS)

LVS was run with `netgen`, comparing the `magic`-extracted device netlist against the `xschem` schematic netlist. The comparison caught two real layout errors before it closed: the input node `Iin` inadvertently shorted to `VSS`, and a signal transistor drawn at $4\times$ its intended width. Both were fixed in the layout. The final run reports *netlists match uniquely*: 10 devices and 7 nets on each side, equivalent cell pin lists (`Iin`, `Vout`, `VDD`, `VSS`), and no property mismatches. The layout is therefore a verified electrical match to the design of record.

4.5 Parasitic Extraction and Post-Layout Simulation

With LVS clean, `magic` extracted a post-layout netlist including parasitic capacitance. This first pass is *cap-only*: it captures coupling and substrate capacitance on every net (105 parasitic C elements) but not wire or gate resistance, which is the standard first pass and models the dominant post-layout effect: the extra node capacitance loading the high-impedance internal nodes. The post-layout testbenches re-inject the same $C_T = 500\ \text{fF}/C_L = 1\ \text{pF}$ loading used pre-layout.

As expected, the parasitics leave the bias untouched: the post-layout operating point settles at $V_{in} = 0.63\ \text{V}$, $V_{out} = 1.67\ \text{V}$, and $I_{tot} = 0.171\ \text{mA}$ ($0.31\ \text{mW}$), tracking the pre-layout sim. The loaded frequency response (the dash-dot trace of Figure 6) holds its DC transimpedance at $59.6\ \text{dB}\Omega$ while the added node capacitance trims the $-3\ \text{dB}$ bandwidth from the folded-device $603\ \text{MHz}$ to $556\ \text{MHz}$ (a further $\sim 8\%$, for a cumulative -12% from the ideal $631\ \text{MHz}$) and lifts peaking to a still-benign $0.22\ \text{dB}$. Table 6 collects the progression. The design is robust to its own layout parasitics: the capacitive-degeneration compensation survives extraction with the flat, well-damped response intact.

Table 6: Loaded response through the physical-implementation flow (SKY130A, TT, 27°C , $C_T = 500\ \text{fF}$, $C_L = 1\ \text{pF}$).

Metric	Ideal ($n_f=1$)	Real devices (folded)	Post-layout (PEX)
DC transimpedance [$\text{dB}\Omega$]	59.4	59.6	59.6
Loaded $f_{-3\text{dB}}$ [MHz]	631	603	556
Peaking [dB]	< 0.1	< 0.1	0.22
Supply current [mA]	0.18	0.18	0.171

Two extraction caveats remain. The pass is cap-only: wire and gate resistance (and hence the

fingering noise benefit of Section 3.8) are not modelled directly, as `magic`'s `extresist` does not converge on the high-resistance poly nets of this cell. Their bandwidth impact is nonetheless bounded to be negligible: inserting a series resistance at the output node (which drives the dominant 1 pF load, making it the most resistance-sensitive node in the circuit) shows that even a pessimistic $50\ \Omega$ of interconnect (well above the few-to-tens of ohms expected on this compact layout) trims the bandwidth by under 1.5% ($556 \rightarrow 548\text{ MHz}$) at unchanged DC gain, and if anything slightly *reduces* peaking by damping the loop. The second caveat: loop phase margin cannot be measured directly on the raw extracted netlist, because the M_1 – M_2 regulation node that the pre-layout testbench breaks for injection is a single physical net in the layout. Checking it requires splitting that node in the extracted deck.

5 Conclusion and Future Work

This report documented the hand design, small-signal analysis, open-source simulation, and physical implementation (schematic capture, layout, LVS, and parasitic extraction) of a regulated-cascode TIA with a capacitive-degeneration bandwidth-extension stage. The original SKY130B baseline (Section 2) was revitalised on SKY130A by discarding the overdrive-voltage sizing in favour of a g_m/I_D methodology (Section 3). The result is a $59.4\text{ dB}\Omega$ transimpedance that, once the CD zero is placed on the ADC-load pole, holds flat out to 631 MHz at only 0.33 mW (a $3\times$ power reduction over the baseline) with $\approx 130^\circ$ of loop phase margin. The design was then laid out in `magic`, closed LVS, and re-simulated with parasitic extraction (Section 4), where the post-layout bandwidth settles at 556 MHz at unchanged gain. The chief remaining limitations are the asymmetric single-ended output swing (Section 3.6) and the sensitivity of the CD compensation to the actual load.

As a visual capstone of that back-end flow, Figure 12 renders the signed-off GDS of Section 4.2 as a full process stack: (a) an *exploded* view that separates the layers vertically (the nwell/diffusion base, the poly and local-interconnect level, and the five metal layers with their connecting vias) and (b) the same stack collapsed into its as-fabricated form. The tight, near-square footprint that motivated the RGC front-end in the first place (Sections 2 and 4.2) is apparent in how few metal levels the routing actually consumes.

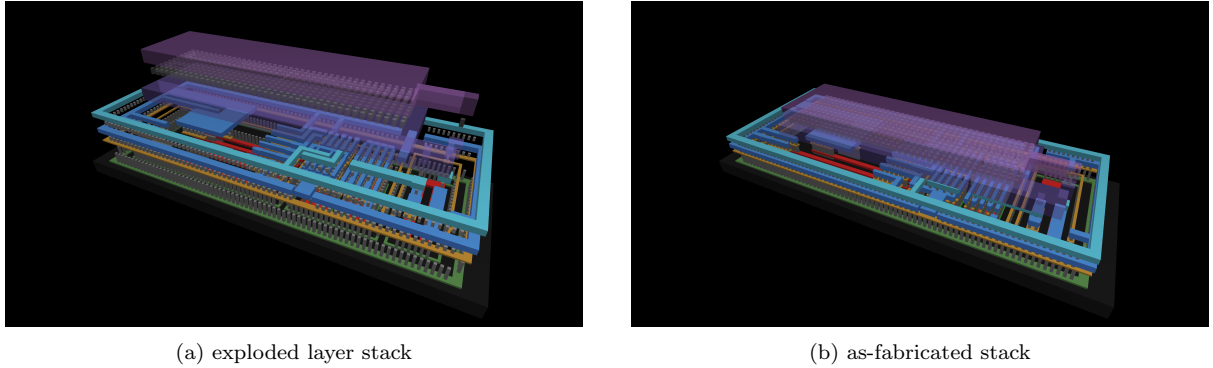


Figure 12: Three-dimensional render of the signed-off RGC-CD TIA GDS: (a) exploded view separating the process layers (diffusion/nwell base through poly, local interconnect, and metal1–metal5), and (b) the collapsed as-fabricated metal stack. Both are the same $25.46 \times 9.8\ \mu\text{m}$ cell of Figure 11.

Future work:

- **Symmetric operation:** bias the output at $\approx V_{DD}/2$ for balanced swing, or adopt a (pseudo-)differential TIA to support true signed bidirectional input currents.
- **Load co-design:** fix the CD zero against the real, extracted current-mirror input and ADC load capacitances so the reported bandwidth reflects the true dominant pole.
- **Process comparison:** port the g_m/I_D design back to SKY130B and A/B the two model sets to quantify the transconductance and parasitic differences.
- **Full extraction:** extend the cap-only PEX of Section 4.5 to a full RC extraction (adding wire and gate resistance, and with it the fingering noise benefit), and re-check the regulation-loop phase margin on the extracted netlist.

A Appendix: Input-Architecture Comparison Study

This study pre-dates the finalised design. It was an early exploration at a higher ~ 1.15 mW operating point with *unloaded* testbenches (no C_T/C_L) and noise integrated over the old 144 MHz band, so its absolute numbers do not match the g_m/I_D design of record of Section 3 (0.33 mW, 59.4 dB Ω , loaded to 631 MHz). In particular its “Baseline” is the original RGC at that operating point, not the finalised converter. What carries over is the *relative* comparison between topologies and the qualitative findings below, not the tabulated absolutes.

To evaluate the biasing, gain-definition and noise trade-offs raised in the conclusion, four TIA input architectures were prototyped in ngspice on SKY130A. All share a 1.8 V supply and a ~ 1.15 mW budget, and use identical testbenches: transimpedance and input impedance are read from a 1 A AC probe at the input ($|V_{out}| = |Z_T|$, $|V_{in}| = |Z_{in}|$), and input-referred current noise is integrated over 1 kHz–144 MHz. The variants are:

- **Baseline:** the original RGC with resistor-divider biasing.
- **A: Shunt-feedback:** the RGC is *removed* and replaced by a single inverting common-source stage (NMOS input, PMOS current-source load) with a feedback resistor R_f from output to input, giving $Z_T \approx -R_f$ and $Z_{in} \approx R_f/(1+A)$.
- **B: Hybrid:** the baseline RGC with an additional *global*, AC-coupled feedback resistor from output back to the input node (nested loops).
- **C: Current-mirror-biased RGC:** the baseline signal path with the input-tail current set by an ideal reference (a proxy for a matched current mirror from a bandgap) in place of R_4 .

Table 7: Architecture comparison (SKY130A, TT, 27°C, ~ 1.15 mW each)

Metric	Baseline	A: Shunt-fb	B: Hybrid	C: CM-bias
Z_T [dB Ω] (Ω)	63.5 (1502)	61.4 (1178)	49.5 (298)	63.7 (1537)
Z_{in} [Ω]	47	322	39	48
Peaking [dB]	0.54	0.02	2.73	0.69
f_{-3dB}	6.9 GHz	5.0 GHz	20 GHz	7.3 GHz
I_{noise} [nA_{rms}]	115	60	234	107
Power [mW]	1.15	1.16	1.15	1.15

Note that the absolute f_{-3dB} values are unrealistically high because these pre-layout testbenches include no input or load capacitance (Section 3.2). The *comparison* between architectures is the meaningful result, not the absolute bandwidth.

A.1 PVT Observations

Across the ss/tt/ff corners the transimpedance of every RGC-based variant moves by only ~ 0.5 dB, because $Z_T \approx R_1$ is set by a resistor rather than by device transconductance. The original design’s *gain* is therefore more PVT-robust than expected. The unregulated quantity is the *bias current*: a $\pm 10\%$ supply sweep (Table 8) moves the total current by +44%, since with resistor loads $I_D \propto V_{DD}$.

Table 8: Total supply current vs. supply voltage ($\pm 10\%$, TT)

Architecture	1.62 V	1.80 V	1.98 V	Spread
Baseline (resistor bias)	0.52 mA	0.64 mA	0.75 mA	+44%
C (tail current mirror)	0.53 mA	0.64 mA	0.75 mA	+42%

Crucially, current-mirror biasing the input tail alone (variant C) barely improves this: the load resistors R_1 – R_3 still set the M2/M3 branch currents, which continue to scale with V_{DD} . In this resistor-loaded topology the load devices do double duty as bias-setting elements, so the bias cannot be robustified without replacing them with current-source loads plus common-mode feedback, which amounts to a different architecture.

A.2 Findings

- **A (shunt-feedback) is the strongest architecture:** the flattest response (0.02 dB peaking), roughly half the input-referred noise (60 vs 115 nA_{rms}), and a transimpedance defined by a single resistor R_f . Its only weakness is a higher input impedance (322 Ω), caused by the modest loop gain of a single-stage forward amplifier ($Z_{in} = R_f/(1 + A)$, $A \approx 4$). **Cascoding the forward stage** raises A and restores a low Z_{in} . Importantly, this forward amplifier is a small (one- to few-transistor) inverting stage, *not* a full operational amplifier, so it preserves the area advantage that motivated the RGC choice.
- **B (hybrid) is counterproductive:** the global loop collapses the transimpedance to 49.5 $dB\Omega$, doubles the noise (234 nA_{rms}), and raises peaking to 2.7 dB. Nested local + global feedback is not worthwhile here. This penalty is fundamental rather than a testbench artefact: the global R_f forms a second shunt path that competes with the existing local RGC regulation, simultaneously subtracting from the forward transimpedance and injecting its own thermal noise at the summing node, so the result holds independent of the load (C_T , C_L) and does not warrant re-evaluation on the loaded RGC-CD.
- **C (CM-biased RGC) $\approx$ baseline:** essentially unchanged signal performance and, as shown above, little bias-robustness gain, a consequence of the resistor-load coupling rather than a flaw in the idea.

References

- [1] N. T. H. Nguyen, I. A. Ukaegbu, J. Sangirov, and M. Hashmi, “Ultra-low power TIA with variable bandwidth in 0.13 μm CMOS for short-range optical interconnects,” *The Journal of Engineering*, vol. 2021, no. 5, pp. 295–300, 2021. [Online]. Available: <https://ietresearch.onlinelibrary.wiley.com/doi/abs/10.1049/tje2.12034>